

DAVID ALVARADO

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Engineer who builds the control layer that makes autonomous delivery safe to run in production — write authority enforced at the tool call, adversarial review gates measured against known-answer benchmarks, and migrations that verify themselves against the live database before they apply. Five years at JPMorgan Chase, building account-opening platforms from scratch and migrating legacy banking systems under zero tolerance for data loss.

TECHNICAL SKILLS

Agentic Systems & Governance: Multi-agent orchestration, policy-as-code governance, agent authority laddering, pre-tool-call enforcement hooks, adversarial review pipelines, LLM evaluation & ground-truth benchmarking, tool calling / function calling, MCP (Model Context Protocol) servers, automated quality & compliance gates

Languages & Frameworks: Java, Spring Boot, Kotlin, Python, JavaScript, SQL, REST APIs, Microservices

Cloud & Infrastructure: AWS, Terraform (Infrastructure as Code), Apache Kafka, Supabase/PostgreSQL, CI/CD

Practices: AI-Native Development, AI Adoption & Enablement, Responsible AI Delivery, Event-Driven Architecture, Legacy Migration & Modernization, Regulated-Environment Delivery, Agile/Scrum

EXPERIENCE

JPMorgan Chase & Co. — Columbus, OH

Software Engineer II — Consumer & Community Banking, Operations Technology

Sept 2021 – Sept 2026

- **Designed and ran a governed multi-agent AI development pipeline** — MCP servers, agent-to-agent invocation, and a written authority model bounding what automation may commit unreviewed, operating inside the firm's mandatory peer review, CI, and change-management controls; sustained across a six-week production rewrite for Legal, Regulatory & Control Operations, since adopted by peers (the same approach hardened independently in Cairn — see Projects).
- **Built an account-opening platform from the ground up as one of six engineers** — owned the data model and service boundaries for a system handling account types too complex for the standard onboarding flow.
- **Migrated a legacy key-entry system to the firm's new banking architecture as one of three engineers** — work integral to Chase's 2023 acquisition of First Republic Bank; owned the reconciliation and cutover strategy for cross-institution account data with zero tolerance for data loss.

Early Careers Workstream Lead (concurrent with engineering role) **Sept 2022 – Sept 2026**

- **Led a 40+ person cross-functional workstream** running the Columbus site's early-careers program across all lines of business — a firm-recognized initiative improving junior-talent recruiting and retention; owned planning, delegation, and stakeholder communication.

PROJECTS

Cairn — Governance Layer for Multi-Agent Delivery — Python, Node.js, Claude Code, MCP *private repository, access on request*

- **Defined a write-authority ladder for autonomous agents** — automation may branch, commit, push, and open a draft pull request; it may not merge, force-push, alter issue state, or delete anything. The ceiling is a draft PR, so a wrong action costs one review rather than a recovery.
- **Moved the ladder from policy to enforcement** — a pre-tool-call hook denies the irreversible actions (PR merge, force-push, repo/issue deletion) outright and prompts before risky ones (branch deletion, issue-body edits); a second hook restricts every agent to the repos in scope. Proven by a refused command, not a passing test.
- **Ran multi-agent pipelines against live repositories** — adversarial code review, cross-repo drift detection, backlog decomposition, and work ranking — 92 recorded dispatches using 1,533 agents and ~204M tokens, each leaving a record of agents completed, tokens spent, and findings confirmed versus refuted. Caught defects human review had missed: a silently reverted configuration block, confirmed by five lenses reading separately, and a merged pull request that left its tracking issue open.
- **Measured the review layer instead of trusting it** — seeded corpora carrying pre-registered ground truth, dispatched blind against the refutation and confirmation stages, returned 12 of 12 matching verdicts; a finding is discarded unless it survives a challenge agent, so the report carries measured signal rather than assertion. Auditing the system with its own pipelines surfaced dead code in the failed-agent guards, where a reviewer that died looked identical to one that found nothing; fixed across all six.

Taskr — Capacity-Fair Household Allocation (PWA) — JavaScript, Supabase/PostgreSQL, Vitest, Vercel [SailorDave17/Taskr](https://github.com/SailorDave17/Taskr)

- **Made an unsafe deployment path safe after an outage** — applying a migration and promoting the client became two deliberate acts in a stated order, so a branch whose client depends on an unapplied migration can no longer reach production; a live-schema probe covers every table, column, RPC, and edge function the client depends on, carrying a negative control so a check reporting success everywhere cannot hide an absence; migrations apply only after the database confirms the byte count and md5 of what it received match the file on disk.
- **Shipped a capacity-proportional allocator** judged against a 13-shape scenario corpus that states plainly when a fair split is arithmetically unreachable rather than approximating one; row-level security and column-level grants enforced server-side and asserted by tests that bypass the client, because a client-side guard is not a guard. A build-time guard refuses any secret-shaped key reaching the client bundle, and a test asserts the guard is asked about every variable the build reads, not the ones someone remembered.

Race Timer — Sailing Start-Sequence Timer — Kotlin, Jetpack Compose, Wear OS / Android [SailorDave17/race-timer](https://github.com/SailorDave17/race-timer)

- **Built a hard-real-time countdown engine** — cut worst-case cue error from ± 212 ms to 61 ms over 150 cues on hardware by rendering cues to PCM and scheduling against a monotonic anchor instead of polling; traced a first-cue latency regression to service-startup ordering and closed it, measured (+172 ms $\rightarrow$ -7 ms). UI-independent Kotlin core unit-tested on the JVM.
- **Shipped it to the store rather than to a README** — GitHub Actions gates every pull request on the test suite and a release build, and a git tag signs and uploads both modules to Google Play; the first build reached internal testers in August 2026. A standalone phone app runs the same timing core rather than a second implementation of the sequence.

EARLIER EXPERIENCE

The Wasserstrom Company — Key Account Manager, Columbus, OH

2016 – 2020

- **Managed \$100M+ in annual business** for Marriott International and Hyatt Hotels, partnering with C-level stakeholders on budgets and multi-year growth.

EDUCATION & CERTIFICATIONS

The Ohio State University — Atmospheric Science & Hospitality Management

We Can Code IT — Software Development Certificate | **US Sailing Level 3 Coach** | **Two-Time National Sailing Champion**